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Roll No

EX-7201 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
High Voltage Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Describe how vacuum breakdown is different from normal breakdown of a gas.
b) Explain the streamer theory of breakdown in air at atmospheric pressure.
2. a) Explain the partial discharge tests on high voltage cables? How is a fault in the insulation located in this test?
b) Explain high voltage tests on Transformer..
3. Explain the different mechanisms by which breakdown occurs in solid dielectrics in practice?
4. Define Townsend's first and second ionisation coefficients. Explain the Townsends criterion for a spark.
5. What are the advantages and limitations of transmitting electrical power at high voltage? What is the need for generating high voltage in laboratory?
6. a) Explain the terms withstand voltage, Flashover voltage, wet and dry power frequency tests as referred to high voltage testing.
b) Describe the significance of impulse tests. Explain the impulse testing of insulators.
7. Explain how and why a sphere gap is used for measurement of high voltage. Discuss the factors that influence the measurement using sphere gap.

OR

Explain the working principle of generating voltmeter with a neat sketch.

8. Explain Paschen's law. What is its significance.
